Propagating Uncertainty Information through Gray-Box Multiobjective Optimization Problems

Work (no longer) in Progress

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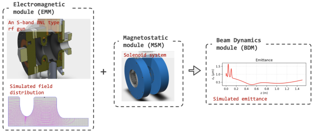
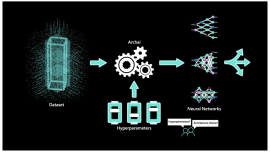
Multiobjective “Simulation” Optimization Problems



Tuning geometry of electron beam RF gun cavity (Linear accelerator) via multi-physics simulations

Automated continuous flow reactor settings for material manufacturing

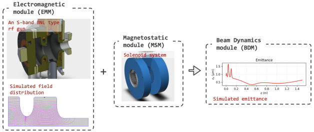
Autotuning (performance optimization) of HPC system libraries



Neural Architecture Search (image credit from Microsoft press release)

Composite "Gray Box" Functions

Sum of squares:



$\rightarrow$ minimize sum-of-squares

One black-box, one simple:

minimize , minimize $f(x)$

$$\min f(x, S(x)) \qquad \text{s.t.} \qquad g(x, S(x)) \leq 0$$

where x are design variables; $S(x)$ is an expensive "simulation" black-box; f and g are algebraic

Surrogate Modeling for Multiobjective Gray Box Optimization

So for MOO, we have an acquisition/scalarization function A, then solve:

$$\min A(f(x, S(x)), w))$$

At Argonne we solve these problems with surrogate based trust-region descent methods.
In each iteration: fit $\hat{S}(x) \approx S(x)$, then solve

$$\min A(f(x, \hat{S}(x)), w)) \qquad \text{s.t.} \qquad x \in [x_0 - r, x_0 + r]$$

parMOO

https://github.com/parmoo/parmoo

Challenge of surrogate uncertainty

Other gray-box tools use Bayesian optimization:

BoTorch

How to do this in ParMOO?

- Need to propagate uncertainty info through multiple layers
- BOTorch does this non intrusively via MC simulation
  - This is not easy, requires advanced MC tools and kernel tricks to do efficiently
- Can we do it analytically?

Option 1: Propagate Uncertainty through nonlinear mappings

Consider:

- A Gaussian process surrogate
- $\rightarrow$ Sum-of-squares function
- $\rightarrow$ Augmented Chebyshev scalarization
- $\rightarrow$ expected improvement acquisition function

For this case, we can propagate the posterior distribution for the GP

- Gaussian distribution
- $\rightarrow$ sum-of-square loss $\rightarrow \chi^2$
- $\rightarrow$ augmented Chebyshev $\rightarrow$ weird piecewise χ^2

Option 2: Solve a constraint optimization problem

Consider:

- A Gaussian process surrogate
- $\rightarrow$ **Any algebraic** function f
- $\rightarrow$ Augmented Chebyshev scalarization
- $\rightarrow$ **Lower confidence bound** acquisition function

For this case, we can directly evaluate the acquisition without propagating

- Evaluate the Gaussian surrogate, get a distribution $S(x) \sim N(\mu_x, \sigma_x)$
- The 95% lower confidence bound would be $\min_x \mu_x - 2\sigma_x$
- For our composite problem, we want to solve $\min_x \text{LCB}(f(x, S(x)))$
- We can solve $\min_x \min_s f(x, s)$ s.t. $s \in [\mu_x - 2\sigma_x, \mu_x + 2\sigma_x]$
  1. Note that we have to solve a nested optimization problem for each function evaluation
  2. So to make it feasible, we need a gradient-based solver
  3. Use the differentiable optimization trick: the gradient $\frac{\delta s}{\delta x}$ = dual solution to the subproblem

TBD

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